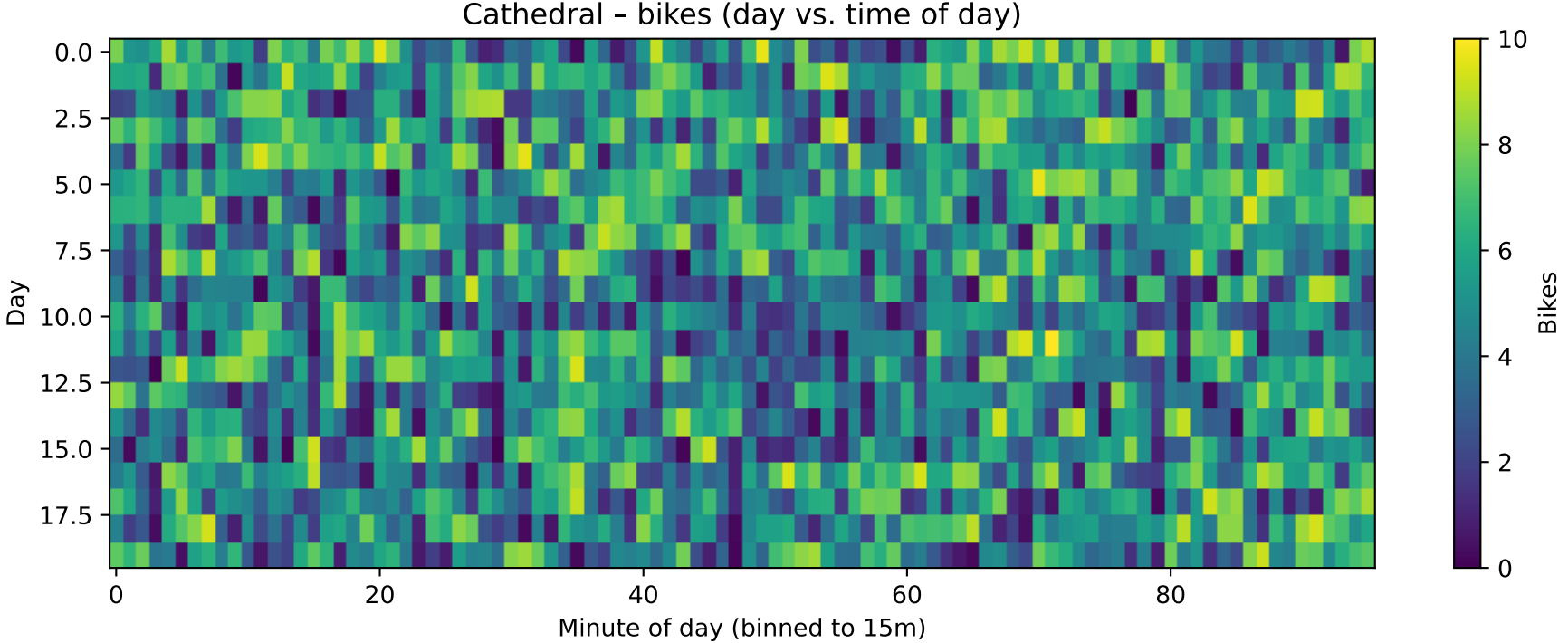
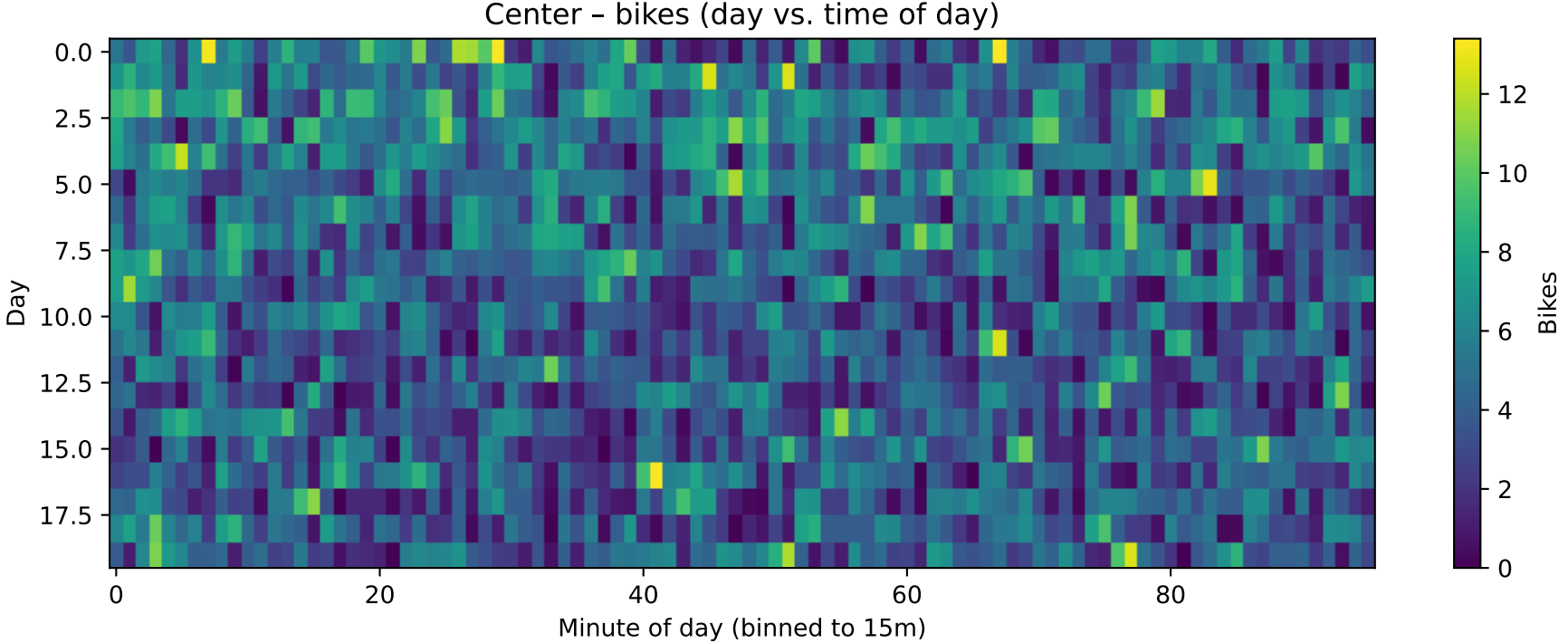
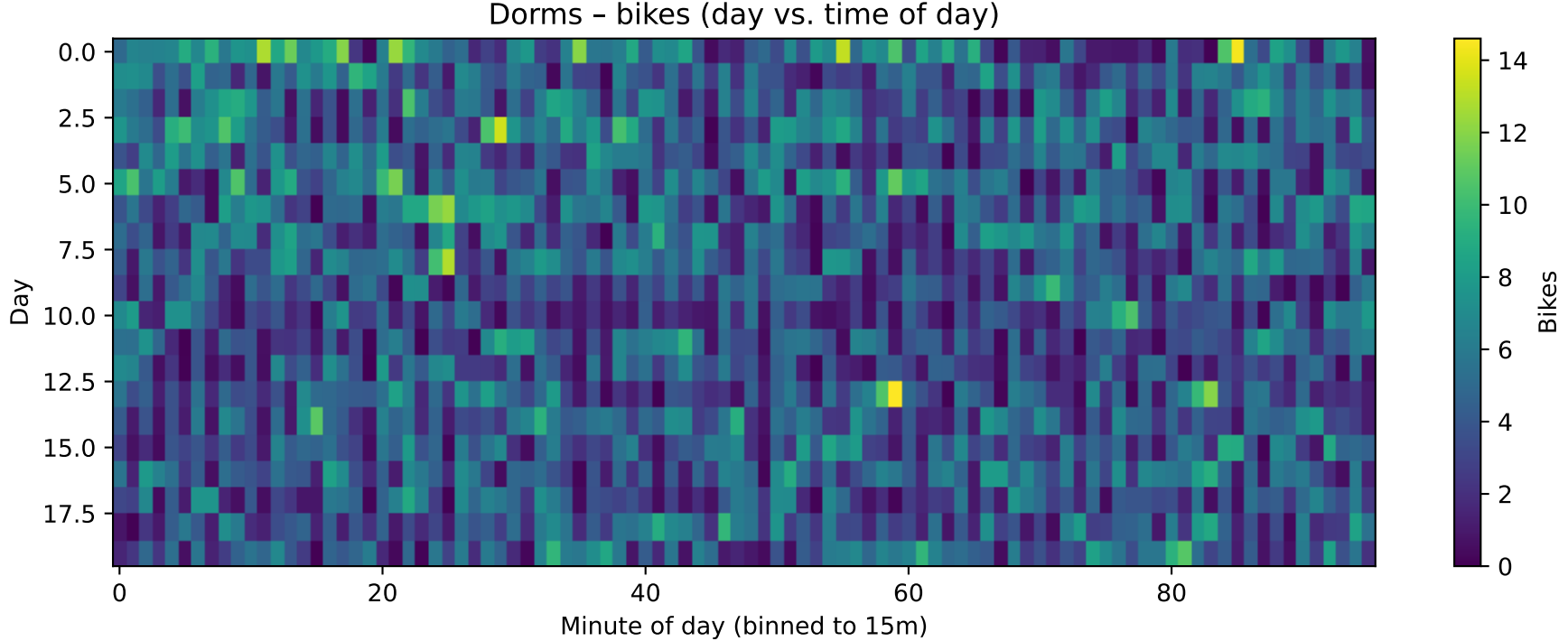
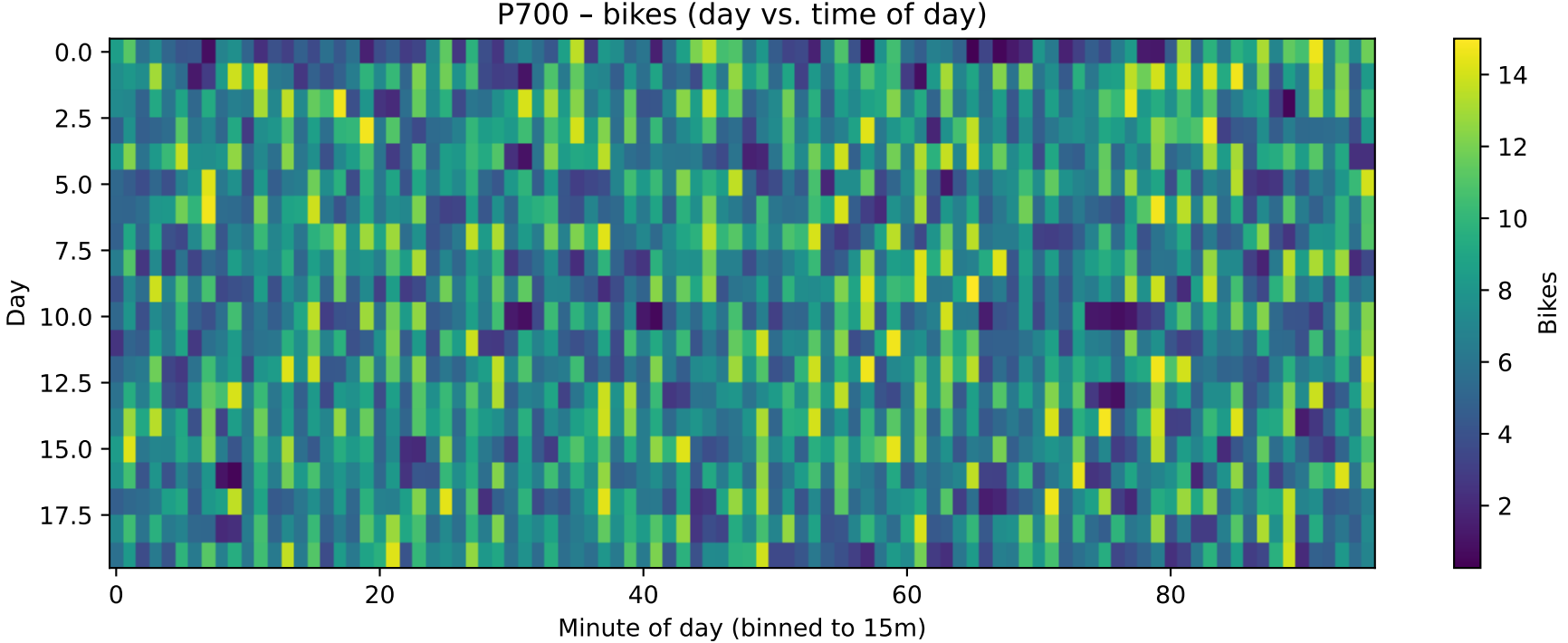
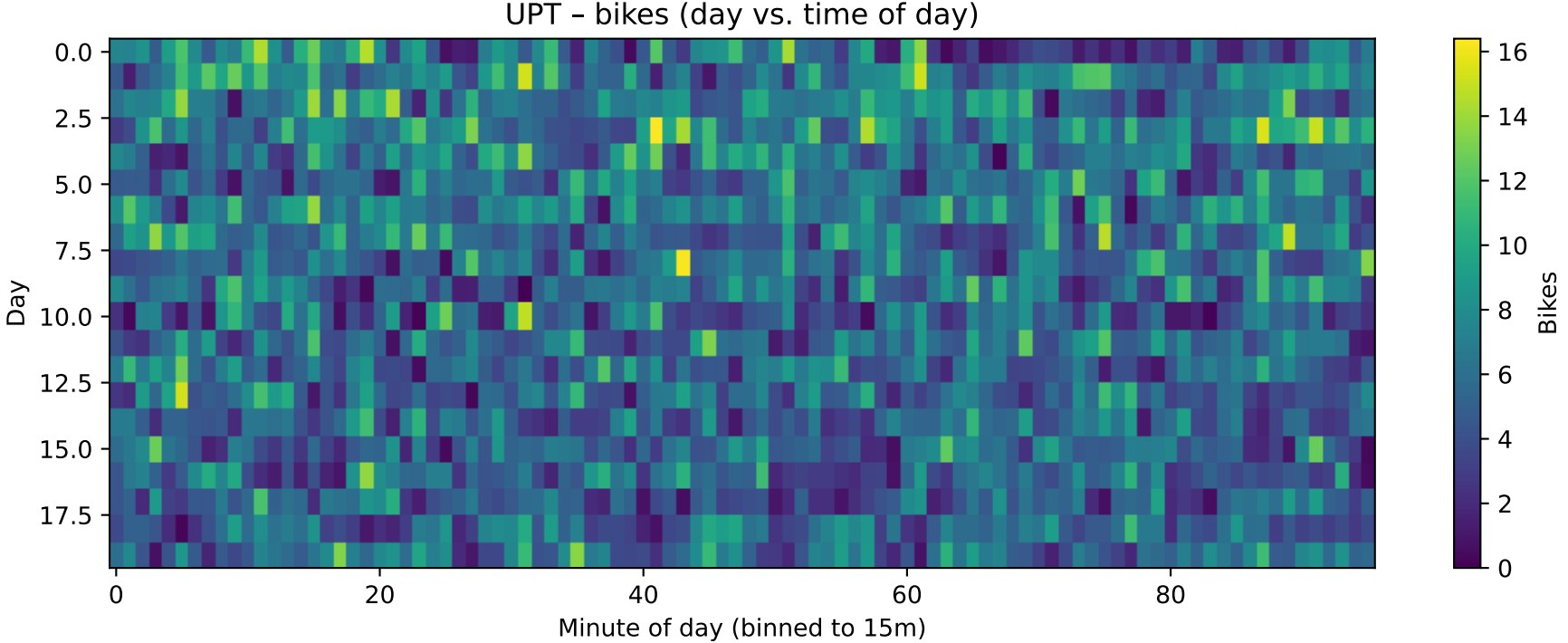
Simulation parameters:
- Duration: 20.0 days (28800 minutes)
- Rebalancing every: 30 minutes (0.020833333333333332 days)
- Breakdown chance: 0.9%
- Repair time: 30 minutes
- Scrapped chance: 0.1%
- Trip time: 2 to 6 minutes
- Probabilities Matrix used: v1

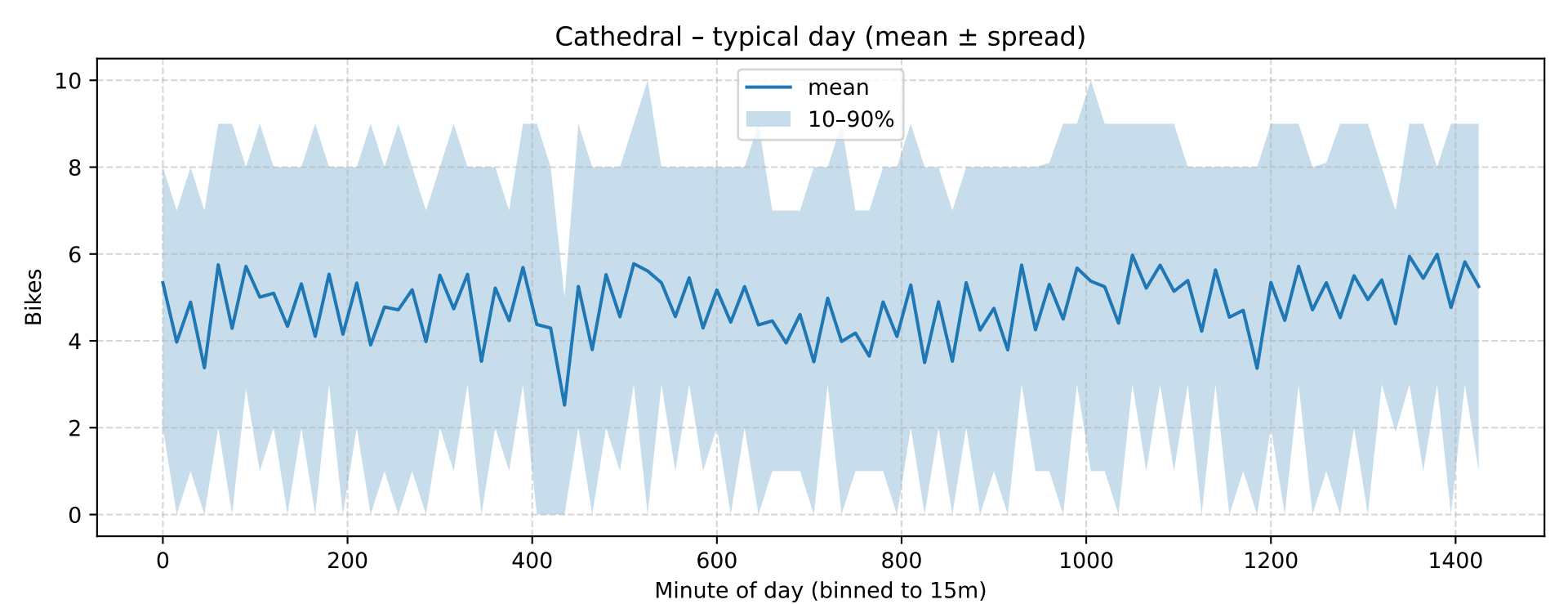

Cathedral – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10

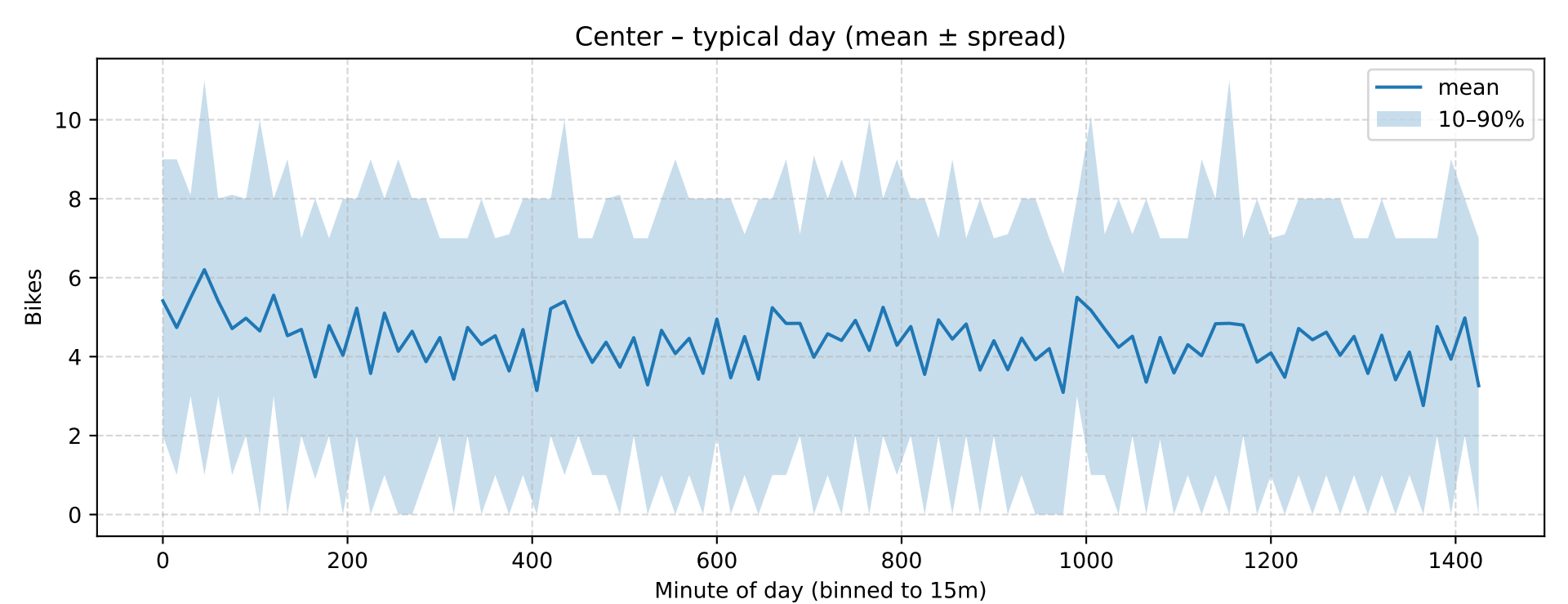

Center – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10
12

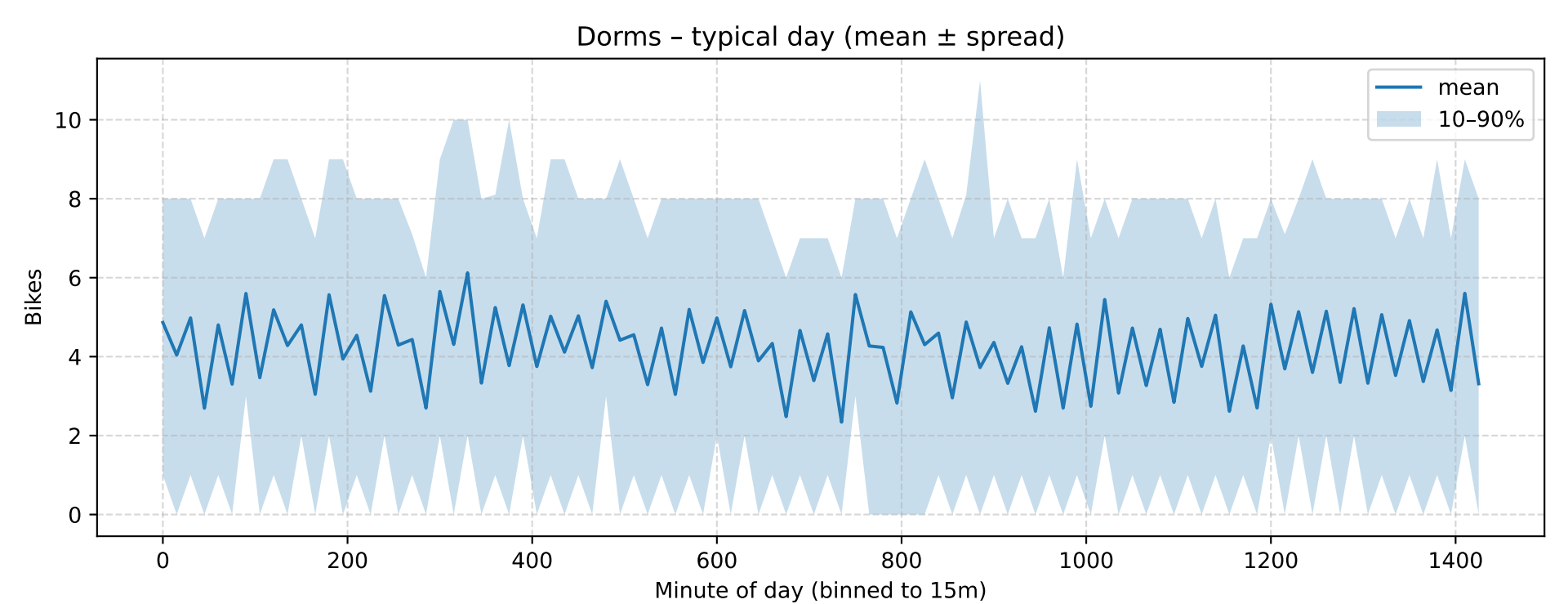

Dorms – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10
12
14

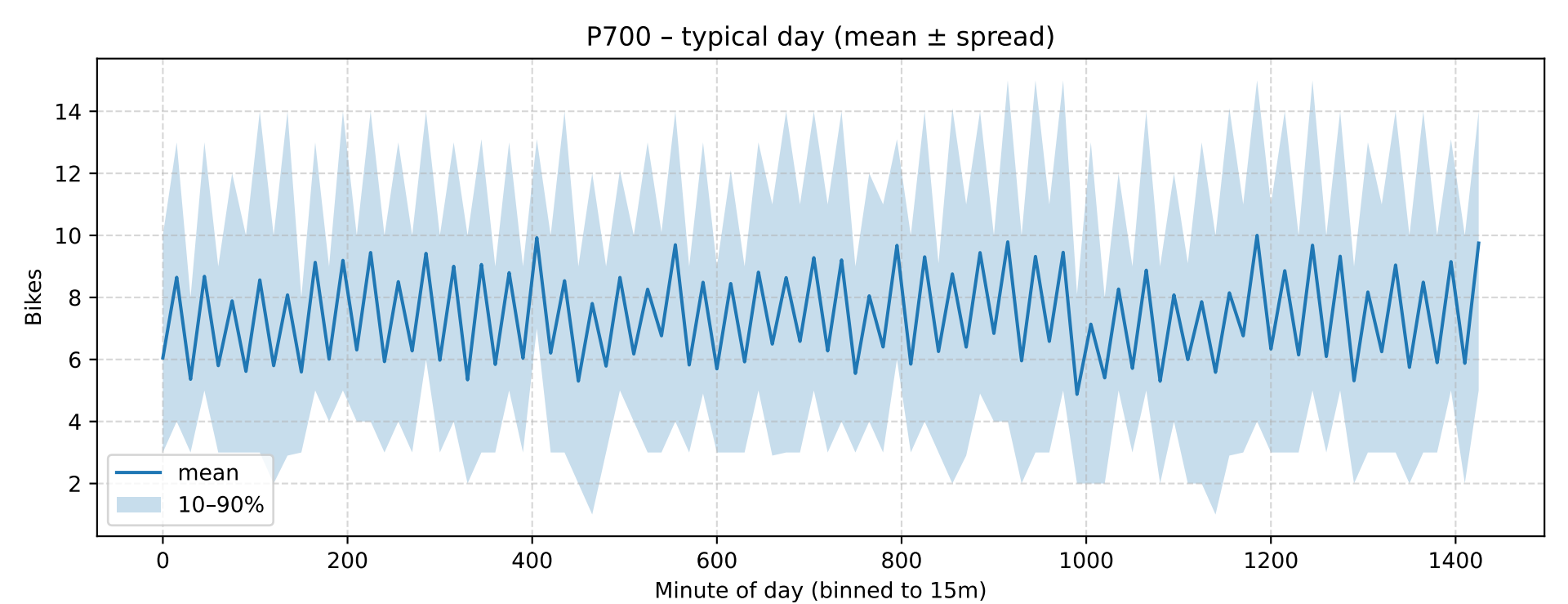

P700 – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
2
4
6
8
10
12
14

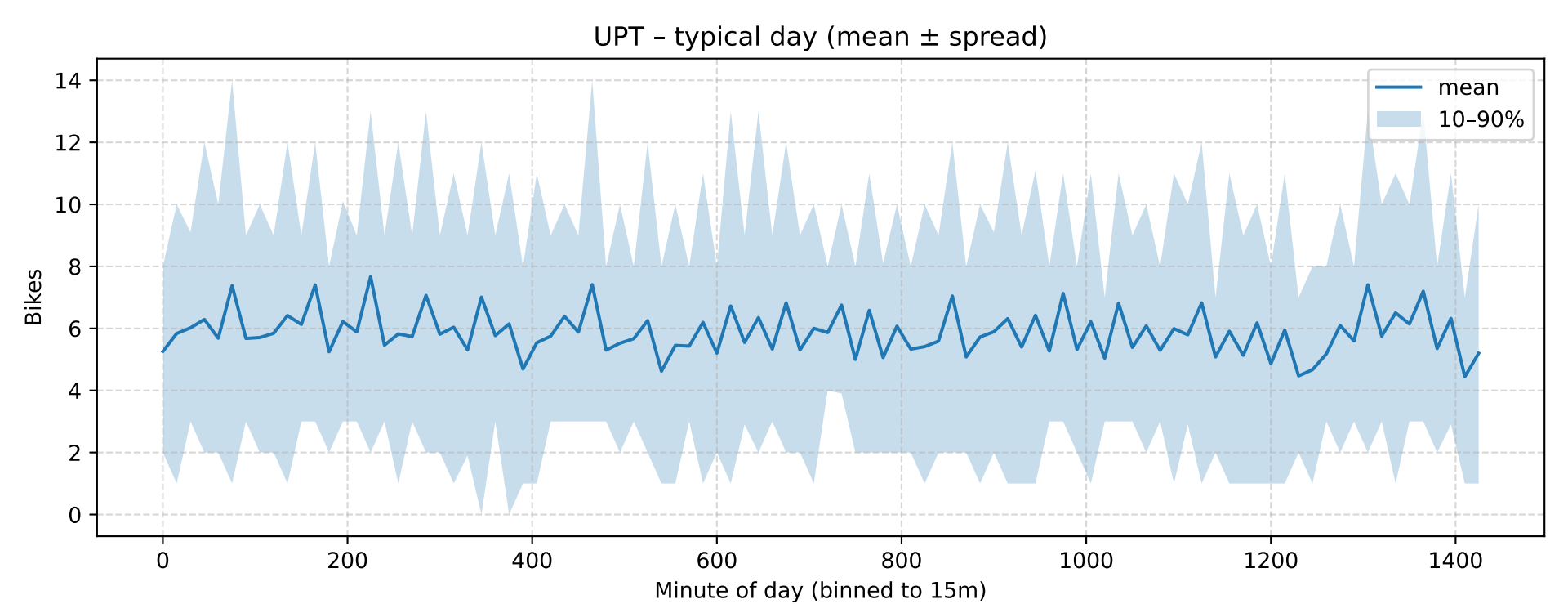

UPT – bikes (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes
0
2
4
6
8
10
12
14
16


Cathedral – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)


Center – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

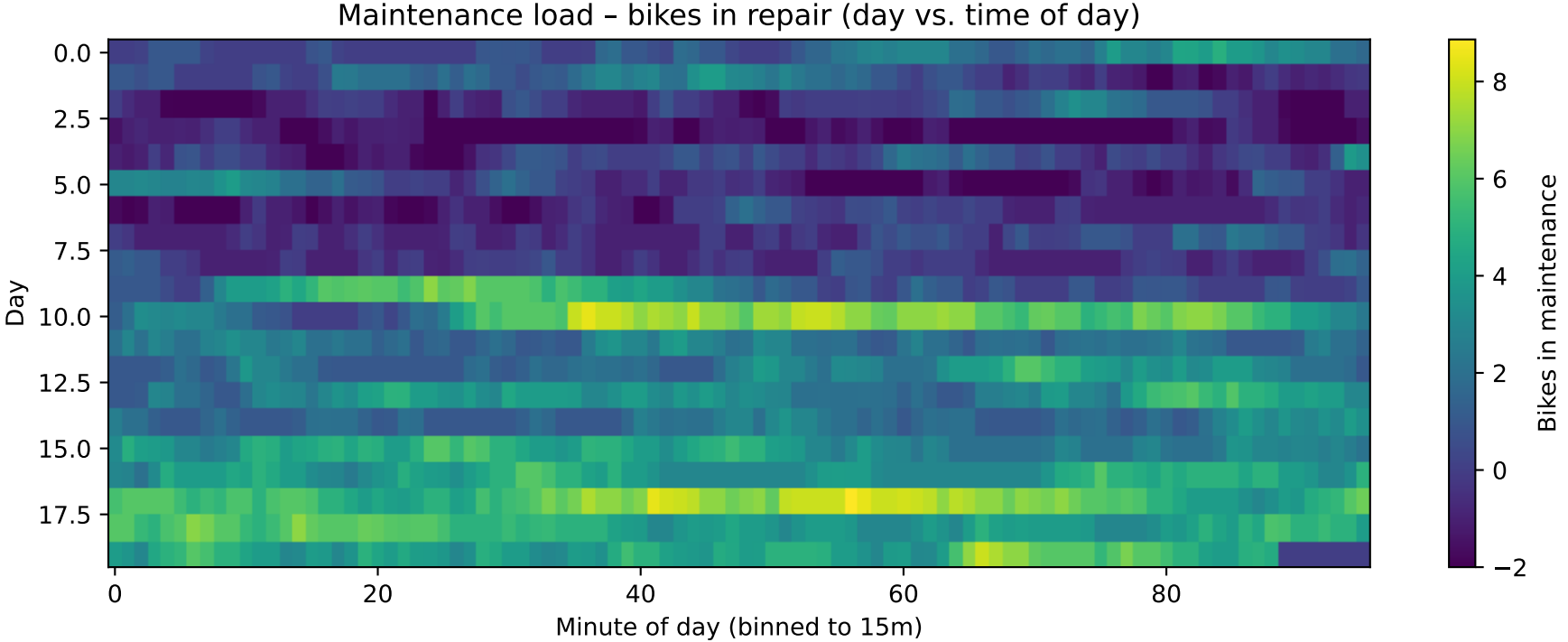

Dorms – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

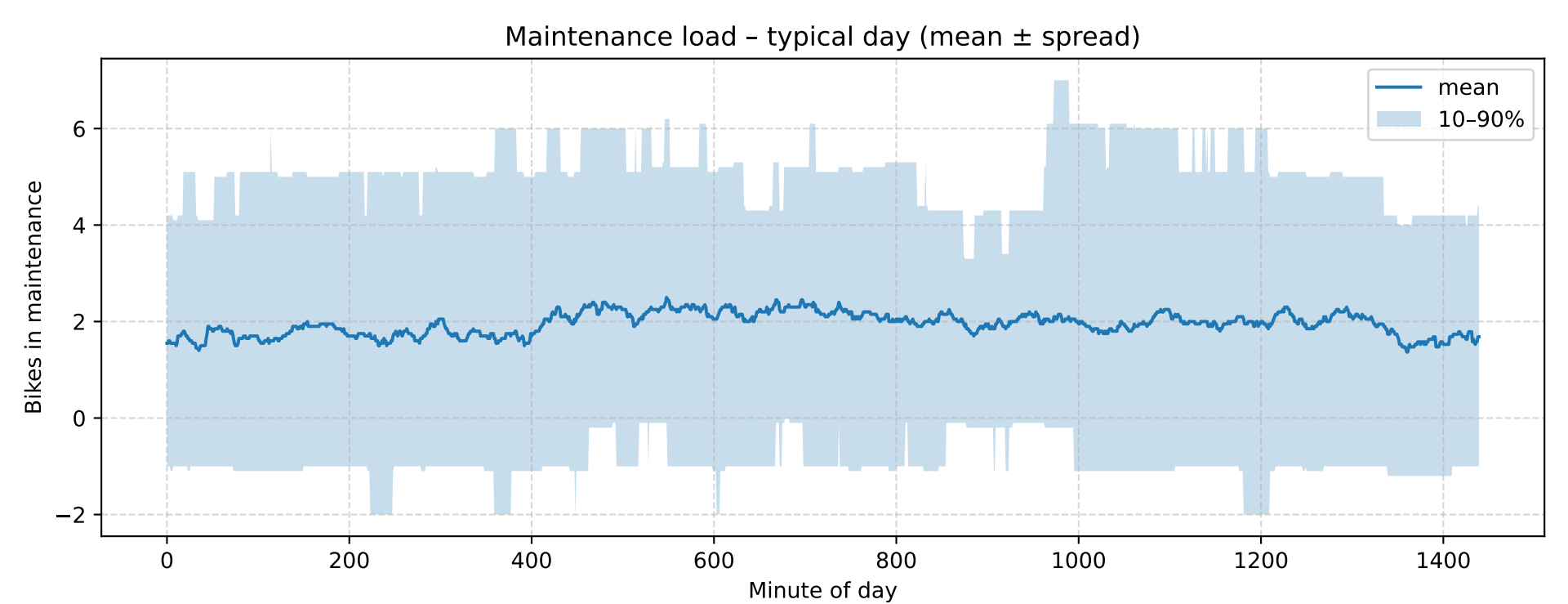

P700 – typical day (mean ± spread)
Bikes
14
12
10
8
6
4
2
mean
10–90%
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

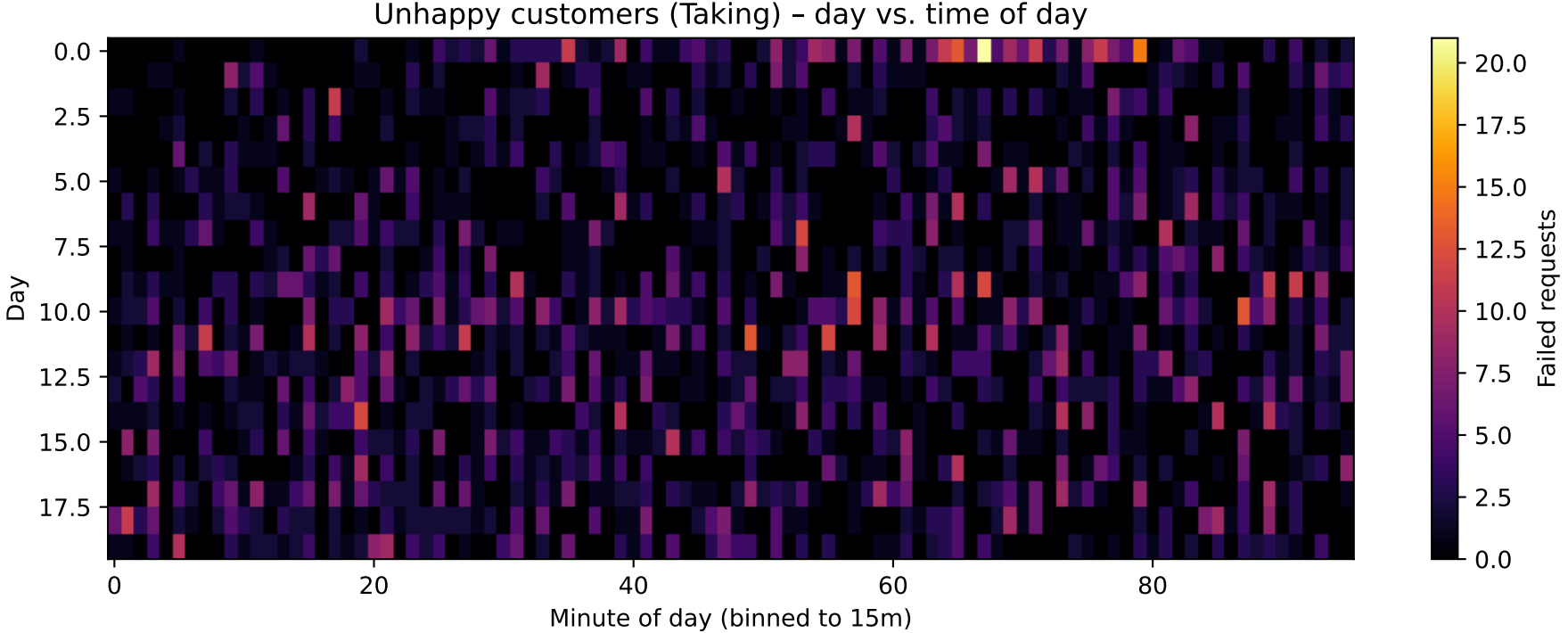

UPT – typical day (mean ± spread)
mean
10–90%
Bikes
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
1400
Minute of day (binned to 15m)

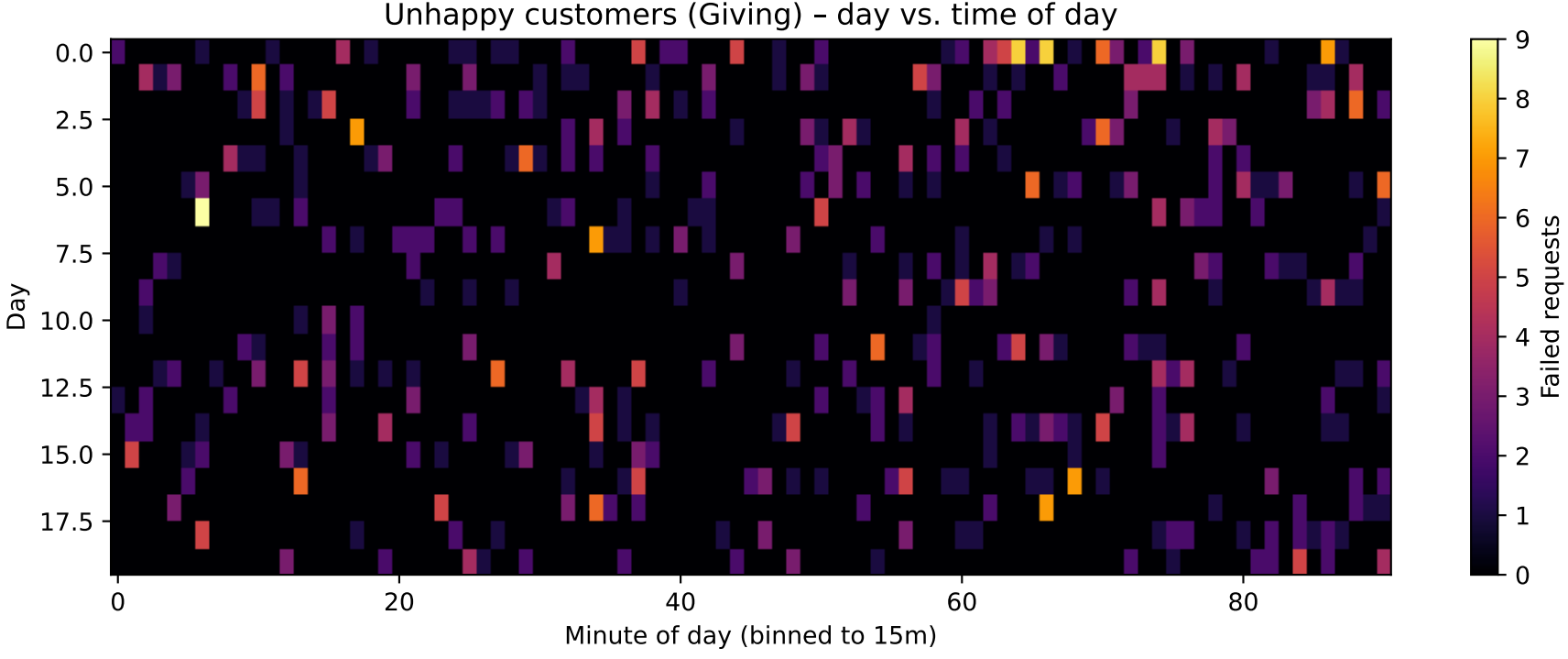
# Station KPIs

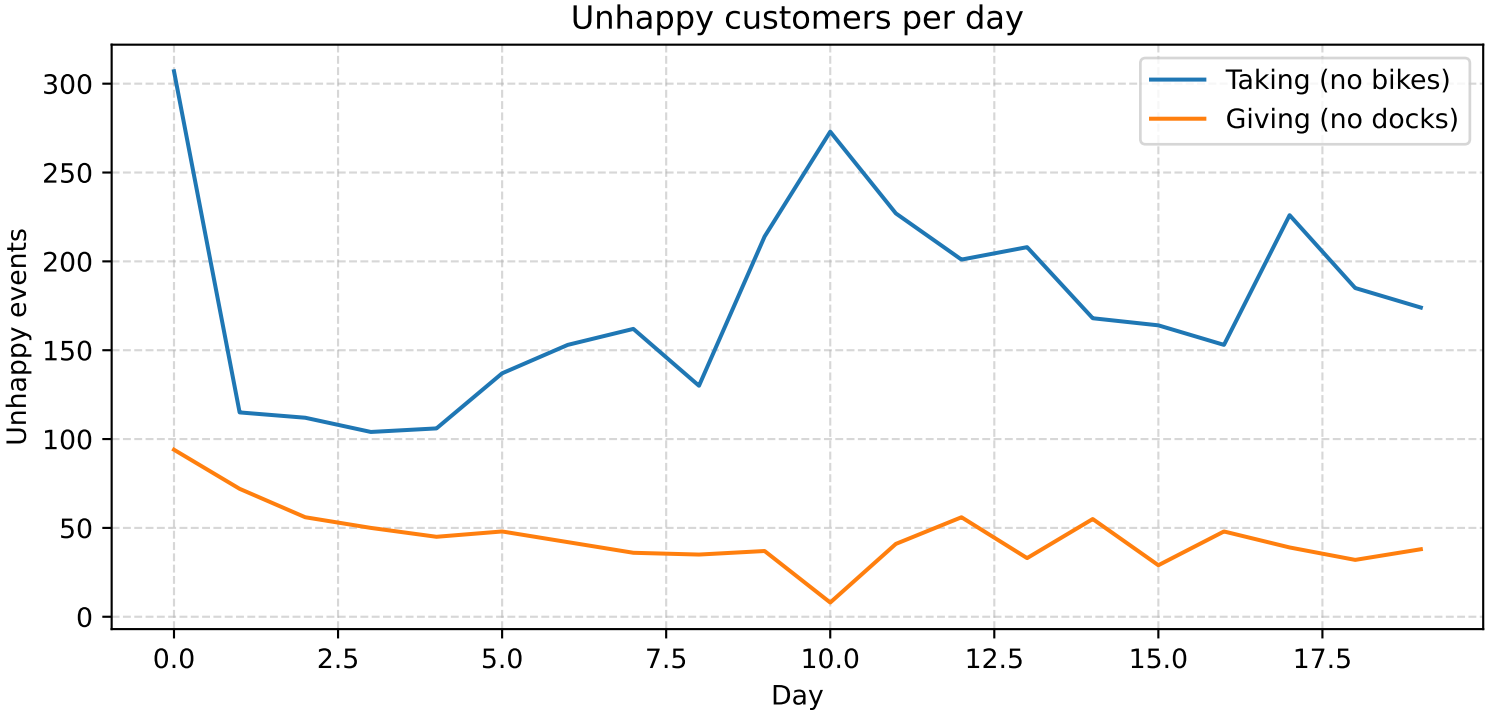
| station | avg_bikes | p10_bikes | p90_bikes | stockout_% | overflow_% |
|---|---|---|---|---|---|
| Dorms | 4.192 | 0.0 | 8.0 | 13.309 | 0.115 |
| Center | 4.376 | 0.0 | 8.0 | 10.368 | 0.083 |
| Cathedral | 4.809 | 1.0 | 8.0 | 7.958 | 3.25 |
| UPT | 5.879 | 2.0 | 10.0 | 3.385 | 0.0 |
| P700 | 7.405 | 3.0 | 13.0 | 1.16 | 3.035 |

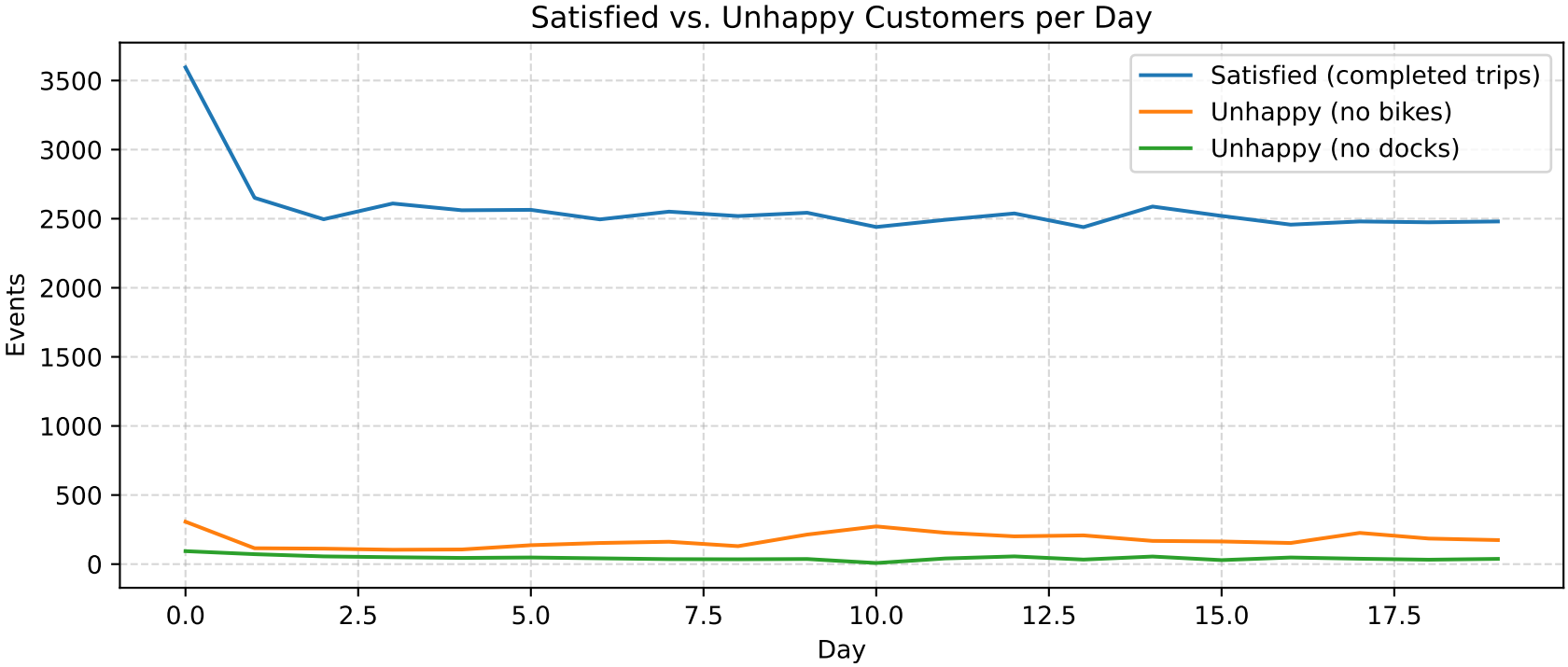

Maintenance load – bikes in repair (day vs. time of day)
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Bikes in maintenance
8
6
4
2
0
−2


Maintenance load – typical day (mean ± spread)
mean
10–90%
Bikes in maintenance
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400
Minute of day


Unhappy customers (Taking) – day vs. time of day
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Failed requests
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0


Unhappy customers (Giving) – day vs. time of day
Day
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
20
40
60
80
Minute of day (binned to 15m)
Failed requests
0
1
2
3
4
5
6
7
8
9


Unhappy customers per day
Taking (no bikes)
Giving (no docks)
Unhappy events
300
250
200
150
100
50
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Day


Satisfied vs. Unhappy Customers per Day
Satisfied (completed trips)
Unhappy (no bikes)
Unhappy (no docks)
Events
3500
3000
2500
2000
1500
1000
500
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Day

Total satisfied trips: 51493
Total unhappy attempts: 4413
Satisfaction ratio: 92.11%
Total scrapped bikes: 5